The documentation and process conversion measures necessary to comply with this revision shall be completed by 1 Feburary 1994

INCH-POUND

MIL-S-19500/454B
1 November 1993
SUPERSEDING
MIL-S-19500/454A
8 August 1983

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON POWER
TYPES 2N5660, 2N5661, 2N5662, AND 2N5663,
JANTX AND JANTXV

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for NPN, silicon, power transistors. Two levels of product assurance are provided for this device type as specified in MIL-S-19500.

1.2 Physical dimensions. For types 2N5660 and 2N5661, see figure 1 (TO-66).
For types 2N5662 and 2N5663, see figure 2 (TO-5).

1.3 Maximum ratings. $T_C$ = +25°C, unless otherwise specified.

| Type | $P_T$ $T_A$ = +25°C | $P_T$ $T_C$ = +100°C | $V_{CBO}$ | $V_{CEO}$ | $V_{EBO}$ | $I_B$ | $I_C$ | $T_J$ and $T_{STG}$ | $V_{CER}$ |
|---|---|---|---|---|---|---|---|---|---|
| | W | W | V dc | V dc | V dc | A dc | A dc | °C | V dc |
| 2N5660 | 2.0 1/ | 20 3/ | 250 | 200 | 6.0 | 0.5 | 2.0 | -65 to +200 | 250 |
| 2N5661 | 2.0 1/ | 20 3/ | 400 | 300 | 6.0 | 0.5 | 2.0 | -65 to +200 | 400 |
| 2N5662 | 1.2 2/ | 15 4/ | 250 | 200 | 6.0 | 0.5 | 2.0 | -65 to +200 | 250 |
| 2N5663 | 1.2 2/ | 15 4/ | 400 | 300 | 6.0 | 0.5 | 2.0 | -65 to +200 | 400 |

1/ Derate linearly, 11.4 mW/°C for $T_A$ > +25°C.
2/ Derate linearly, 6.9 mW/°C for $T_A$ > +25°C.
3/ Derate linearly, 200 mW/°C for $T_C$ > +100°C.
4/ Derate linearly, 150 mW/°C for $T_C$ > +100°C.

1.4 Primary electrical characteristics.

| Limits | $h_{FE2}$ 1/ $I_C$ = 0.5 A dc, $V_{CE}$ = 5 V dc | | $\lvert h_{fe} \rvert$ $I_C$ = 0.1 A dc, $V_{CE}$ = 5 V dc, f = 10 MHz | $V_{BE(sat)}$ 1/ $I_C$ = 1.0 A dc, $I_B$ = 0.1 A dc | $V_{CE(sat)}$ 1/ $I_C$ = 1 A dc, $I_B$ = 0.1 A dc | Pulse response | | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | $t_{on}$ $I_C$ = 0.5 A dc | $t_{off}$ $I_C$ = 0.5 A dc | |
| | 2N5660 2N5662 | 2N5661 2N5663 | | | | | 2N5660 2N5662 | 2N5661 2N5663 |
| | | | | V dc | V dc | µs | µs | µs |
| Minimum | 40 | 25 | 2 | | | | | |
| Maximum | 120 | 75 | 7 | 1.2 | 0.4 | 0.25 | 0.85 | 1.2 |

1/ Pulsed.
$R_{\theta JC}$ for 2N5660, 2N5661 = 5 °C/W maximum; for 2N5662, 2N5663 = 6.67 °C/W maximum.
$R_{\theta JA}$ for 2N5660, 2N5661 = 87.5 °C/W maximum; for 2N5662, 2N5663 = 145.8 °C/W maximum.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: NASA/Parts Project Office (NPPO), NASA Goddard Space Flight Center, Code 310.A, Greenbelt, MD 20771 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5961

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated detail specification. The individual item requirements shall be in accordance with MIL-S-19500 and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-S-19500.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500 and figures 1 and 2 herein.

3.3.1 Lead material and finish. Lead material for 2N5662 and 2N5663 shall be Kovar or Alloy 52. Lead finish shall be gold plated, or solder dipped. For 2N5661 and 2N5661 lead finish shall be solderable in accordance with MIL-S-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4 Marking. Marking shall be in accordance with MIL-S-19500.



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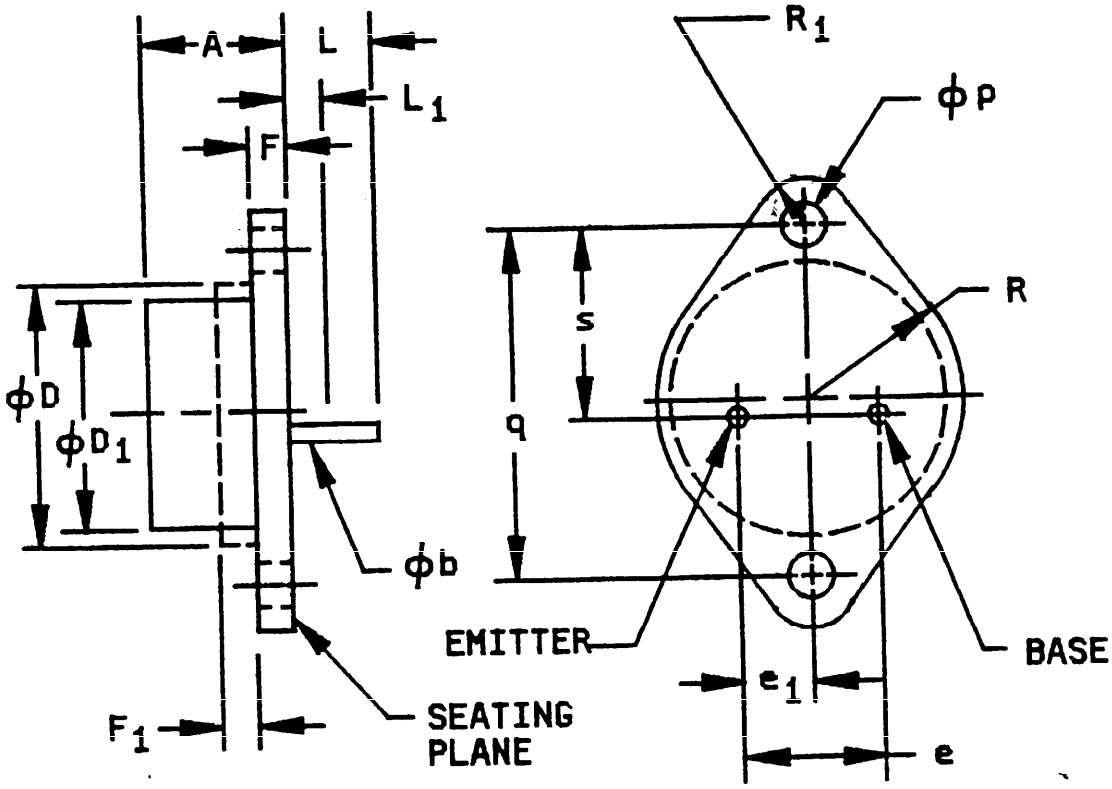



FIGURE 1. Physical dimensions for 2N5660 and 2N5661 (TO-66).



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| Symbol | Dimensions | | | | Notes |
|---|---|---|---|---|---|
| | Inches | | Millimeters | | |
| | Min | Max | Min | Max | |
| A | .250 | .340 | 6.35 | 8.64 | |
| $\phi b$ | .028 | .034 | 0.71 | 0.86 | 5,9 |
| $\phi D$ | | .620 | | 15.75 | 3 |
| $\phi D_1$ | .470 | .500 | 11.94 | 12.70 | 3 |
| e | .190 | .210 | 4.83 | 5.33 | 4 |
| $e_1$ | .093 | .107 | 2.36 | 2.72 | 4 |
| F | .050 | .075 | 1.27 | 1.91 | 3 |
| $F_1$ | | .050 | | 1.27 | 3 |
| L | .360 | .500 | 9.14 | 12.70 | 5 |
| $L_1$ | | .050 | | 1.27 | 5,9 |
| $\phi p$ | .142 | .152 | 3.61 | 3.86 | 7 |
| q | .958 | .962 | 24.33 | 24.43 | |
| R | | .350 | | 8.89 | |
| $R_1$ | .115 | .145 | 2.92 | 3.68 | 6 |
| s | .570 | .590 | 14.48 | 14.99 | 4 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Body contour is optional within zone defined by $\phi D$, $\phi D_1$, $F_1$.
4. These dimensions should be measured at points .050 inch (1.27 mm) to .055 inch (1.40 mm) below seating plane. When gauge is not used, measurement will be made at seating plane.
5. Both terminals.
6. At both ends.
7. Two holes.
8. The collector shall be electrically connected to the case.
9. $\phi b$ applies between $L_1$ and L. Diameter is uncontrolled in $L_1$.

FIGURE 1. Physical dimensions for 2N5660 and 2N5661 (TO-66) - Continued.



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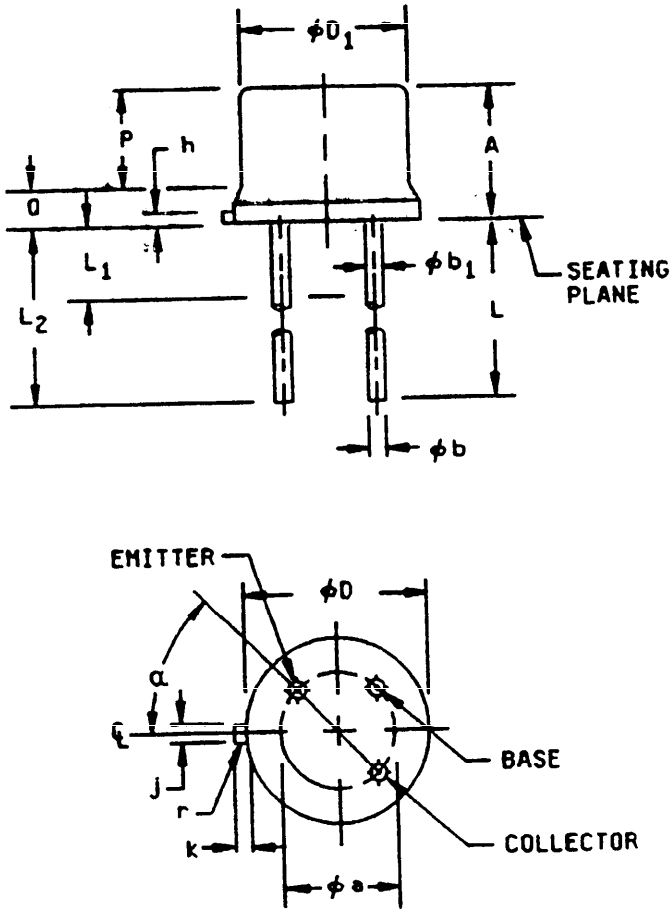



FIGURE 2. Physical dimensions for 2N5662 and 2N5663 (TO-5).



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| Symbol | Dimensions | | | | Notes |
|---|---|---|---|---|---|
| | Inches | | Millimeters | | |
| | Min | Max | Min | Max | |
| A | .240 | .260 | 6.10 | 6.60 | |
| $\phi a$ | 200 TP | | 5.080 TP | | 7 |
| $\phi b$ | .016 | .021 | .407 | 0.533 | 8, 9 |
| $\phi b_1$ | .016 | .019 | .407 | 0.482 | 8, 9 |
| $\phi D$ | .335 | .370 | 8.51 | 9.40 | |
| $\phi D_1$ | .305 | .335 | 7.75 | 8.51 | |
| h | .009 | .041 | .23 | 1.04 | |
| j | .028 | .034 | .712 | .863 | 3 |
| k | .029 | .045 | .74 | 1.14 | 4 |
| L | 1.500 | 1.750 | 38.10 | 44.45 | 8, 9 |
| $L_1$ | | .050 | | 1.27 | 8, 9 |
| $L_2$ | .250 | | 6.35 | | 8, 9 |
| P | .100 | | 2.54 | | 6 |
| Q | | .050 | | 1.27 | 5 |
| r | | .010 | | .254 | 11 |
| $\alpha$ | 45° TP | | 45° TP | | 7 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Beyond r (radius) maximum, J shall be held for a minimum length of 0.011 inch (0.28 mm).
4. K measured from maximum $\phi D$.
5. Outline in this zone is not controlled.
6. $\phi D_1$ shall not vary more than 0.010 inch (0.25 mm) in zone P. This zone is controlled for automatic handling.
7. Leads at gauge plane 0.054 +0.001 - .000 inches (1.37 +0.03 - .00 mm) below seating plane shall be within 0.007 inch (0.18 mm) radius of true position (TP) at maximum material condition (MMC) relative to tab at MMC. The device may be measured by direct methods or by the gauge and gauge procedures shown in figure 3.
8. $\phi b_1$ applies between $L_1$ and $L_2$. $\phi b$ applies between $L_2$ and L minimum. Diameter is uncontrolled in $L_1$ and beyond L minimum.
9. All three leads.
10. The collector shall be electrically connected to the case.
11. r (radius) applies to both inside corners of tab.

FIGURE 2. Physical dimensions for 2N5662 and 2N5663 (TO-5) - Continued.



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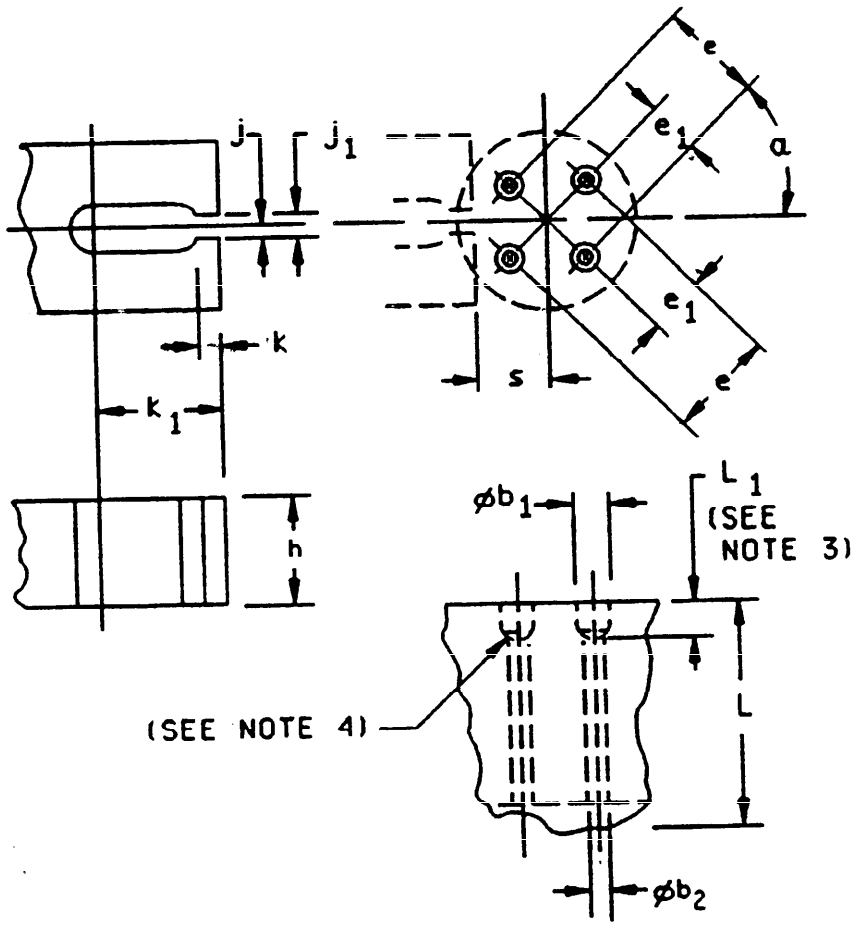



FIGURE 3. Gauge for lead and tab location for types 2N5662 and 2N5663(TO-5).



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| Symbol | Dimensions | | | | Notes |
|---|---|---|---|---|---|
| | Inches | | Millimeters | | |
| | Min | Max | Min | Max | |
| $\phi b_1$ | 0.0595 | 0.0605 | 1.511 | 1.537 | |
| $\phi b_2$ | 0.0325 | 0.0335 | 0.855 | 0.851 | |
| e | 0.1995 | 0.2005 | 5.067 | 5.093 | |
| $e_1$ | 0.0995 | 0.1005 | 2.527 | 2.552 | |
| h | 0.150 Nominal | | 3.81 Nominal | | |
| j | 0.0175 | 0.0180 | 0.4445 | 0.457 | |
| $j_1$ | 0.035 | 0.0355 | 0.889 | 0.902 | |
| k | 0.009 | 0.011 | 0.23 | 0.28 | |
| $k_1$ | 0.125 Nominal | | 3.18 Nominal | | |
| L | 0.372 | 0.378 | 9.45 | 9.60 | |
| $L_1$ | 0.054 | 0.055 | 1.37 | 1.40 | |
| s | 0.182 | 0.199 | 4.62 | 5.05 | 3 |
| $\alpha$ | 44.90° | 45.10° | 44.90° | 45.10° | |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The location of the tab locator within the limits indicated will be determined by the tab and flange dimensions of the device being checked.
4. The following gauging procedure shall be used:
   a. The device being measured shall be inserted until its seating plane is 0.125 inch (3.18 mm) ± 0.010 inch (0.25 mm) from the seating surface of the gauge. A force of 8 ± 0.5 ounces shall then be applied parallel and symmetrical to the device's cylindrical axis. When examined visually after the force application (the force need not be removed) the seating plane of the device shall be seated against the gauge.
   b. The use of a pin straightener prior to insertion in the gauge is permitted.
5. Gauging plane.
6. Drill angle.

FIGURE 3. Gauge for lead and tab location for types 2N5662 and 2N5663 (TO-5) - Continued.



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4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500 and as specified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500.

4.3 Screening (JANTX and JANTXV levels). Screening shall be in accordance with table II of MIL-S-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

| Screen (see table II of MIL-S-19500) | Measurement<br>JANTX and JANTXV levels |
|---|---|
| 1/ | Thermal response (see 4.3.2) |
| 9 | $I_{CES1}$ |
| 11 | $I_{CES1}$ and $h_{FE2}$; $\Delta I_{CES1}$ = 100 percent of initial value or 20 nA dc, whichever is greater. |
| 12 | See 4.3.1 |
| 13 | Subgroup 2 of table I herein;<br>$\Delta I_{CES1}$ = 100 percent of initial value or 20 nA dc;<br>$\Delta h_{FE}$ = ±25 percent of initial value. |

1/ This test shall be performed anytime before screen 9.

4.3.1 Power burn-in. Power burn-in conditions are as follows:

$T_J$ = 187.5 ±12.5°C; $V_{CE}$ = 100 V dc; $T_A \leq +100°C$

NOTE: No heat sink or forced air cooling on the devices shall be permitted.

4.3.2 Thermal response ($\Delta V_{BE}$ measurements). The $\Delta V_{BE}$ measurements shall be performed in accordance with MIL-STD-750, method 3131. The $\Delta V_{BE}$ conditions ($I_H$ and $V_H$) and maximum limit $V_{BE}$ shall be derived by each vendor. The chosen $\Delta V_{BE}$ measurement and conditions for each device in the qualification lot shall be submitted in the qualification report and a thermal response curve shall be plotted, read, and measurements recorded. The chosen $\Delta V_{BE}$ shall be considered final after the manufacturer has had the opportunity to test five consecutive lots. The following parameter measurements shall apply:

a. $I_H$ measurement . . . . . 10 mA.

b. $V_{CE}$ measurement voltage . . . . . 10 V (same as $V_H$).

c. $I_H$ collector heating current . . . . . 1.0 A (minimum).

d. $V_H$ collector-emitter heating voltage . . . . . 10 V (minimum).

e. $t_H$ heating time . . . . . 10 ms (TO-5).
. . . . . 25 ms (TO-66).

f. $t_{MD}$ measurement delay time . . . . . 50 µs to 80 µs.



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4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500 and table I herein. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps of table III herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVb (JANTX and JANTXV) of MIL-S-19500, and as follows. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps of table III herein.

4.4.2.1 Group B inspection, table IVb of MIL-S-19500.

a. Subgroup 3: Steady-state operation life; method 1027, $T_J$ = 187.5°C ±12.5°C; $V_{CE}$ = 100 V dc; $T_A \leq$ +100°C; (see 4.3.1).

b. Subgroup 7: Safe operating area (switching); method 3053; load condition C (unclamped inductive load) (see figure 8); $T_C$ = +100°C; duty cycle ≤ 2 percent; $R_S$ = 0.62 ohms; $V_{CC}$ = 25 V dc; $t_p$ = 50 µs (vary to obtain $I_C$); $V_{BB1}$ = 10 V dc; $V_{BB2}$ = 0.5 V dc; $R_{BB1}$ = 20 ohms; $R_{BB2}$ = 100 ohms; $I_C$ = 2 A dc; L = 0.1 mH, .072 ohms (26 turns, 1 row of #20 AWG wire on Arnold Engineering Company Core #A-930157-2, or equivalent).

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps of table III herein.

4.4.3.1 Group C inspection, table V of MIL-S-19500.

a. Subgroup 2:

(1) Thermal shock (glass strain); method 1056; test condition B.

(2) Terminal strength (tension) for types 2N5660 and 2N5661; method 2036; test condition A; weight = 3 pounds; t = 15s.

(3) Terminal strength (lead fatigue) for types 2N5662 and 2N5663; method 2036; test condition E.

b. Subgroup 6:

(1) Steady-state operation life for 2N5660 and 2N5661; method 1026; $T_C$ = +100°C; $P_T$ = 20 W; $V_{CE}$ = 20 V dc.

(2) Steady-state operation life for 2N5662 and 2N5663; method 1026; $T_A$ = +25°C; $P_T$ = 1.2 W; $V_{CE}$ = 40 V dc.

4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VII of MIL-S-19500, and table II herein. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table III herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.



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TABLE I. Group A inspection.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 1 | | | | | | |
| Visual and mechanical examination | 2071 | | | | | |
| Subgroup 2 | | | | | | |
| Breakdown voltage, collector to emitter | 3011 | Pulsed (see 4.5.1), $I_C$ = 10 mA dc, bias condition D | $V_{(BR)CEO}$ | | | V dc |
| 2N5660, 2N5662 | | | | 200 | | |
| 2N5661, 2N5663 | | | | 300 | | |
| Breakdown voltage, collector to base | 3011 | Bias condition D, $I_C$ = 10 mA dc, pulsed (see 4.5.1), $R_{BE}$ = 100 ohms | $V_{(BR)CER}$ | | | V dc |
| 2N5660, 2N5662 | | | | 250 | | |
| 2N5661, 2N5663 | | | | 400 | | |
| Breakdown voltage, emitter to base | 3026 | Bias condition D, $I_E$ = 10 µA dc | $V_{(BR)EBO}$ | 6 | | V dc |
| Collector to emitter cutoff current | 3041 | Bias condition C | $I_{CES1}$ | | | |
| 2N5660, 2N5662 | | $V_{CE}$ = 200 V dc | | | 0.2 | µA dc |
| 2N5661, 2N5663 | | $V_{CE}$ = 300 V dc | | | 0.2 | µA dc |
| Collector to base cutoff current | 3036 | Bias condition D | $I_{CBO}$ | | | |
| 2N5660, 2N5662 | | $V_{CB}$ = 200 V dc | | | 0.1 | µA dc |
| | | $V_{CB}$ = 250 V dc | | | 1.0 | mA dc |
| 2N5661, 2N5663 | | $V_{CB}$ = 300 V dc | | | 0.1 | µA dc |
| | | $V_{CB}$ = 400 V dc | | | 1.0 | mA dc |
| Base emitter voltage (saturated) | 3066 | Test condition A, pulsed (see 4.5.1), $I_C$ = 1 A dc, $I_B$ = 0.1 A dc | $V_{BE(sat1)}$ | | 1.2 | V dc |
| Base emitter voltage (saturated) | 3066 | Test condition A, pulsed (see 4.5.1), $I_C$ = 2 A dc, $I_B$ = 0.4 A dc | $V_{BE(sat2)}$ | | 1.5 | V dc |
| Collector to emitter voltage (saturated) | 3071 | Pulsed (see 4.5.1), $I_C$ = 1 A dc, $I_B$ = 0.1 A dc | $V_{CE(sat1)}$ | | 0.4 | V dc |
| Collector to emitter voltage (saturated) | 3071 | Pulsed (see 4.5.1), $I_C$ = 2 A dc, $I_B$ = 0.4 A dc | $V_{CE(sat2)}$ | | 0.8 | V dc |

See footnote at end of table.



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TABLE I. Group A inspection - Continued.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 2 - Continued | | | | | | |
| Forward current transfer ratio | 3076 | $V_{CE}$ = 2.0 V dc, $I_C$ = 50 mA dc, pulsed (see 4.5.1) | $h_{FE1}$ | | | |
| 2N5660, 2N5662 | | | | 40 | | |
| 2N5661, 2N5663 | | | | 25 | | |
| Forward current transfer ratio | 3076 | $V_{CE}$ = 5.0 V dc, $I_C$ = 0.5 A dc, pulsed (see 4.5.1) | $h_{FE2}$ | | | |
| 2N5660, 2N5662 | | | | 40 | 120 | |
| 2N5661, 2N5663 | | | | 25 | 75 | |
| Forward current transfer ratio | 3076 | $V_{CE}$ = 5.0 V dc, $I_C$ = 1 A dc, pulsed (see 4.5.1) | $h_{FE3}$ | 15 | | |
| Forward current transfer ratio | 3076 | $V_{CE}$ = 5.0 V dc, $I_C$ = 2 A dc, pulsed (see 4.5.1) | $h_{FE4}$ | 5 | | |
| Subgroup 3 | | | | | | |
| High temperature operation: | | $T_A$ = +150°C | | | | |
| Collector to emitter cutoff current | 3041 | Bias condition C | $I_{CES2}$ | | | |
| 2N5660, 2N5662 | | $V_{CE}$ = 200 V dc | | | 100 | µA dc |
| 2N5661, 2N5663 | | $V_{CE}$ = 300 V dc | | | 100 | µA dc |
| Low temperature operation | | $T_A$ = -55°C | | | | |
| Forward current transfer ratio | 3076 | $V_{CE}$ = 5.0 V dc, $I_C$ = 0.5 A dc, pulsed (see 4.5.1) | $h_{FE5}$ | | | |
| 2N5660, 2N5662 | | | | 15 | | |
| 2N5661, 2N5663 | | | | 10 | | |

See footnote at end of table.



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TABLE I. Group A inspection - Continued.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 4 | | | | | | |
| Small-signal short-circuit forward-current transfer ratio | 3306 | $V_{CE}$ = 5 V dc, $I_C$ = 0.1 A dc, f = 10 MHz | $\|h_{fe}\|$ | 2 | 7 | |
| Pulse response | | | | | | |
| Turn-on time | 3251 | Test condition A, $V_{CC}$ = 100 V dc, $I_C$ = 0.5 A dc | $t_{on}$ | | | |
| 2N5660, 2N5662 | | See figure 4 | | | 0.25 | µs |
| 2N5661, 2N5663 | | See figure 5 | | | 0.25 | µs |
| Turn-off time | 3251 | Test condition A, $V_{CC}$ = 100 V dc, $I_C$ = 0.5 A dc | $t_{off}$ | | | |
| 2N5660, 2N5662 | | See figure 4 | | | 0.85 | µs |
| 2N5661, 2N5663 | | See figure 5 | | | 1.2 | µs |
| Open circuit output capacitance | 3236 | $V_{CB}$ = 10 V dc, $I_E$ = 0, 100 kHz ≤ f ≤ 1 MHz | $C_{obo}$ | | 45 | pF |
| Subgroup 5 | | | | | | |
| Safe operating area (continuous dc) | 3051 | $T_C$ = +100°C, power application time ≥ 1 second, 1 cycle, $t_r + t_f$ = 10 µs, (2N5660 and 2N5661, see figure 6, 2N5662 and 2N5663, see figure 7) | | | | |
| Test 1 | | | | | | |
| 2N5660, 2N5661 | | $V_{CE}$ = 10 V dc, $I_C$ = 2 A dc | | | | |
| 2N5662, 2N5663 | | $V_{CE}$ = 7.5 V dc, $I_C$ = 2 A dc | | | | |
| Test 2 | | | | | | |
| 2N5660, 2N5661 | | $V_{CE}$ = 40 V dc, $I_C$ = 500 mA dc | | | | |
| 2N5662, 2N5663 | | $V_{CE}$ = 25 V dc, $I_C$ = 600 mA dc | | | | |
| Test 3 | | | | | | |
| 2N5660 | | $V_{CE}$ = 200 V dc, $I_C$ = 36 mA dc | | | | |
| 2N5662 | | $V_{CE}$ = 200 V dc, $I_C$ = 27 mA dc | | | | |
| Test 4 | | | | | | |
| 2N5661 | | $V_{CE}$ = 300 V dc, $I_C$ = 19 mA dc | | | | |
| 2N5663 | | $V_{CE}$ = 300 V dc, $I_C$ = 14 mA dc | | | | |
| Safe operating area (clamped inductive) | | $T_C$ = +100°C, $t_r + t_f$ ≤ 10 µs duty cycle ≤ 2 percent, $I_C$ = 2 A dc, $t_p$ ≈ 4 ms, $V_{CE}$ = 25 V dc (see figures 9 and 10) | | | | |
| 2N5660, 2N5662 | | Clamp voltage = 200, +0, -5 V dc | | | | |
| 2N5661, 2N5663 | | Clamp voltage = 300, +0, -5 V dc | | | | |
| Electrical measurements | | See table III, steps 1 and 4 | | | | |

1/ For sampling plan, see MIL-S-19500.



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TABLE II. Group E inspection (all quality levels) for qualification only.

| Inspection | MIL-STD-750 | | Qualification conformance inspection |
|---|---|---|---|
| | Method | Conditions | |
| Subgroup 1 | | | 22 devices, c = 0 |
| Thermal shock (temperature cycling) | 1051 | 500 cycles, condition D | |
| Electrical measurements | | See table III, steps 1, 2, 3, 4, 6, 7, and 8 | |
| Subgroup 2 | | | 22 devices, c = 0 |
| Steady-state dc Blocking life | 1039 or 1049 | Condition A, 500 hours | |
| Electrical measurements | | See table III, steps 1, 6, 7, and 8 | |
| Subgroup 3 | | | |
| Not applicable | | | |
| Subgroup 4 | | | 22 devices, c = 0 |
| Thermal resistance | 3131 | $R_{\theta JC}$ = 5°C/W for 2N5660 and 2N5661 maximum, $R_{\theta JC}$ = 6.67°C/W for 2N5662 and 2N5663 maximum (see 4.5.2) | |
| Subgroup 5 | | | 15 devices, c = 0 |
| Barometric pressure (reduced) | 1001 | Normal mounting pressure = 8 mm Hg ±2 mm Hg for 60 s (minimum) | |
| Electrical measurements | | See table III, step 5. | |



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TABLE III. Groups A, B, C, and E electrical end-point measurements. 1/ 2/

| Step | Inspection | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|---|
| | | Method | Conditions | | Min | Max | |
| 1. | Collector to emitter cutoff current | 3041 | Bias condition C | $I_{CES1}$ | | | |
| | 2N5660, 2N5662 | | $V_{CE}$ = 200 V dc, | | | 0.2 | µA dc |
| | 2N5661, 2N5663 | | $V_{CE}$ = 300 V dc | | | 0.2 | µA dc |
| 2. | Collector to emitter voltage (saturated) | 3071 | $I_C$ = 1 A dc, $I_B$ = 0.1 A dc, (pulsed, see 4.5.1) | $V_{CE(sat1)}$ | | 0.4 | V dc |
| 3. | Base emitter voltage (saturated) | 3066 | Test condition A, $I_C$ = 1 A dc, $I_B$ = 0.1 A dc, (pulsed, see 4.5.1) | $V_{BE(sat1)}$ | | 1.2 | V dc |
| 4. | Forward current transfer ratio | 3076 | $V_{CE}$ = 5 V dc, $I_C$ = 0.5 A dc, (pulsed, see 4.5.1) | $h_{FE2}$ | | | |
| | 2N5660, 2N5662 | | | | 40 | 120 | |
| | 2N5661, 2N5663 | | | | 25 | 75 | |
| 5. | Collector to base cutoff current | 3036 | Bias condition D | $I_{CBO}$ | | | |
| | 2N5660, 2N5661 | | $V_{CB}$ = 250 V dc | | | 1 | V dc |
| | 2N5662, 2N5663 | | $V_{CB}$ = 400 V dc | | | 1 | V dc |
| 6. | Collector to emitter cutoff current | 3041 | Bias condition C | $\Delta I_{CES1}$ | 100 percent of initial value or 20 nA dc, whichever is greater. | | |
| | 2N5660, 2N5662 | | $V_{CE}$ = 200 V dc | | | | |
| | 2N5661, 2N5663 | | $V_{CE}$ = 300 V dc | | | | |
| 7. | Forward current transfer ratio | 3076 | $V_{CE}$ = 5 V dc, $I_C$ = 0.5 A dc, pulsed (see 4.5.1) | $\Delta h_{FE2}$ | ±25 percent change in initial recorded value. | | |
| 8. | Thermal resistance | 3131 | See 4.5.2 | $R_{\theta JC}$ | | | °C/W |

1/ The electrical measurements for table IVb (JANTX and JANTXV) of MIL-S-19500 are as follows:

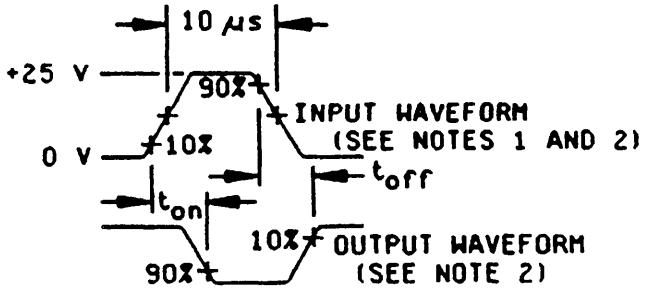
a. Subgroup 2, see table III herein, steps 1 and 3.
b. Subgroup 3, see table III herein, steps 1, 2, 3, 4, 6, 7, and 8.
c. Subgroup 6, see table III herein, steps 1, 4, 6, and 7.

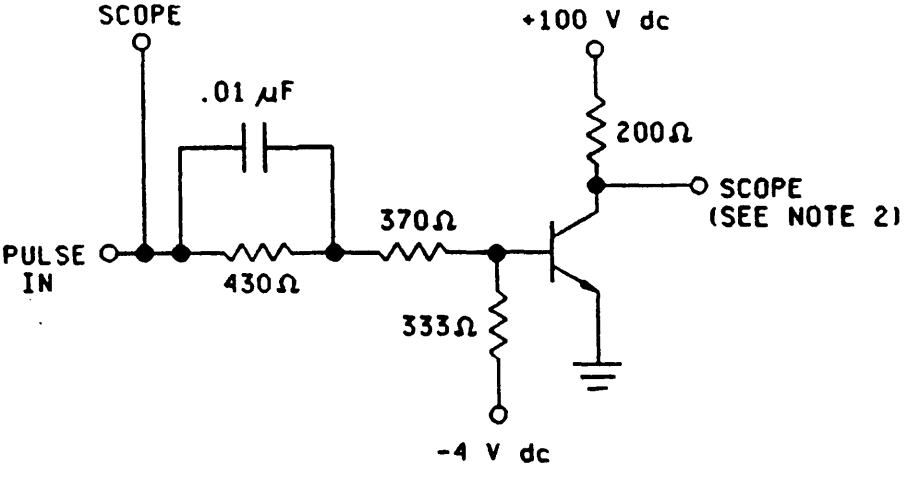
2/ The electrical measurements for table V of MIL-S-19500 are as follows:

a. Subgroup 2, see table III herein, steps 1, 2, and 4.
b. Subgroup 3, see table III herein, steps 1, 2, and 4.
c. Subgroup 6, see table III herein, steps 1, 2, 3, 4, 6, and 7.



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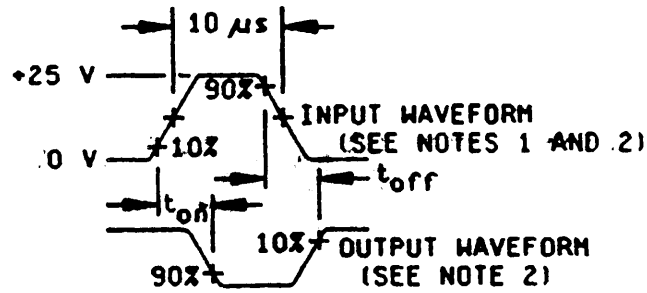



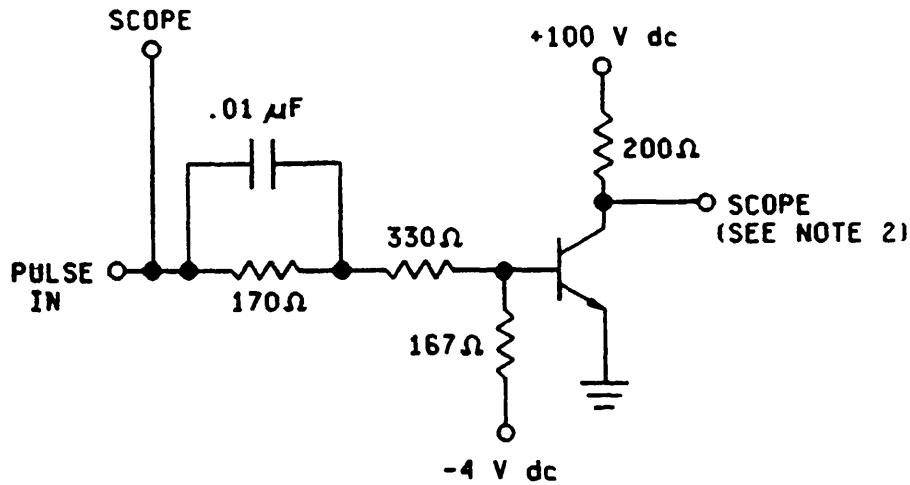

NOTES:
1. The input waveform is supplied by a pulse generator with the following characteristics: $t_r \leq 15$ ns, $t_f \leq 15$ ns, $Z_{OUT} = 50$ ohms, PW = 10 µs, duty cycle ≤ 2 percent.
2. Output waveforms are monitored on an oscilloscope with the following characteristics: $t_r \leq 1$ ns, $Z_{IN} \geq 10$ megohms, $C_{IN} \leq 11.5$ pF.
3. Resistors shall be noninductive types.
4. The dc power supplies may require additional bypassing in order to minimize ringing.
5. The input pulse voltages, -4 V dc and +100 V dc, are nominal and shall be adjusted to obtain $I_{B1} = -I_{B2} = 15$ mA dc.
6. The 0.01 µF capacitor may be removed for current adjustment only.

FIGURE 4. Pulse response test circuit for types 2N5660 and 2N5662.



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NOTES:
1. The input waveform is supplied by a pulse generator with the following characteristics: $t_r \leq 15$ ns, $t_f \leq 15$ ns, $Z_{OUT} = 50$ ohms, PW = 10 µs, duty cycle ≤ 2 percent.
2. Output waveforms are monitored on an oscilloscope with the following characteristics: $t_r \leq 1$ ns, $Z_{IN} \geq 10$ megohms, $C_{IN} \leq 11.5$ pF.
3. Resistors shall be noninductive types.
4. The dc power supplies may require additional by=passing in order to minimize ringing.
5. The input pulse voltages, -4 V dc and +100 V dc, are nominal and shall be adjusted to obtain $I_{B1} = -I_{B2} = 25$ mA dc.
6. The 0.01 µF capacitor may be removed for current adjustment only.

FIGURE 5. Pulse response test circuit for types 2N5661 and 2N5663.



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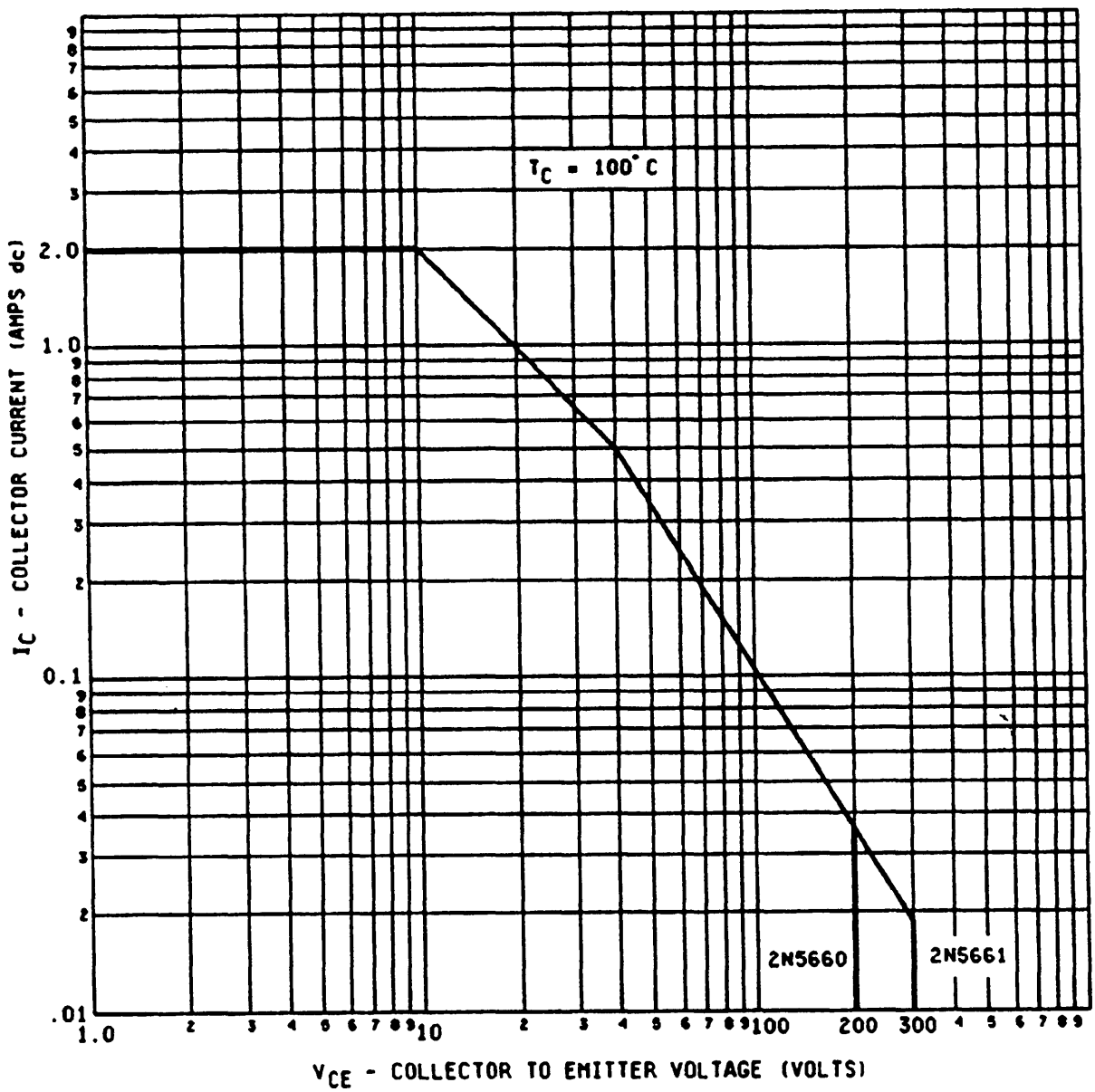



FIGURE 6. Maximum operating area graph (continuous dc) for types 2N5660 and 2N5661.



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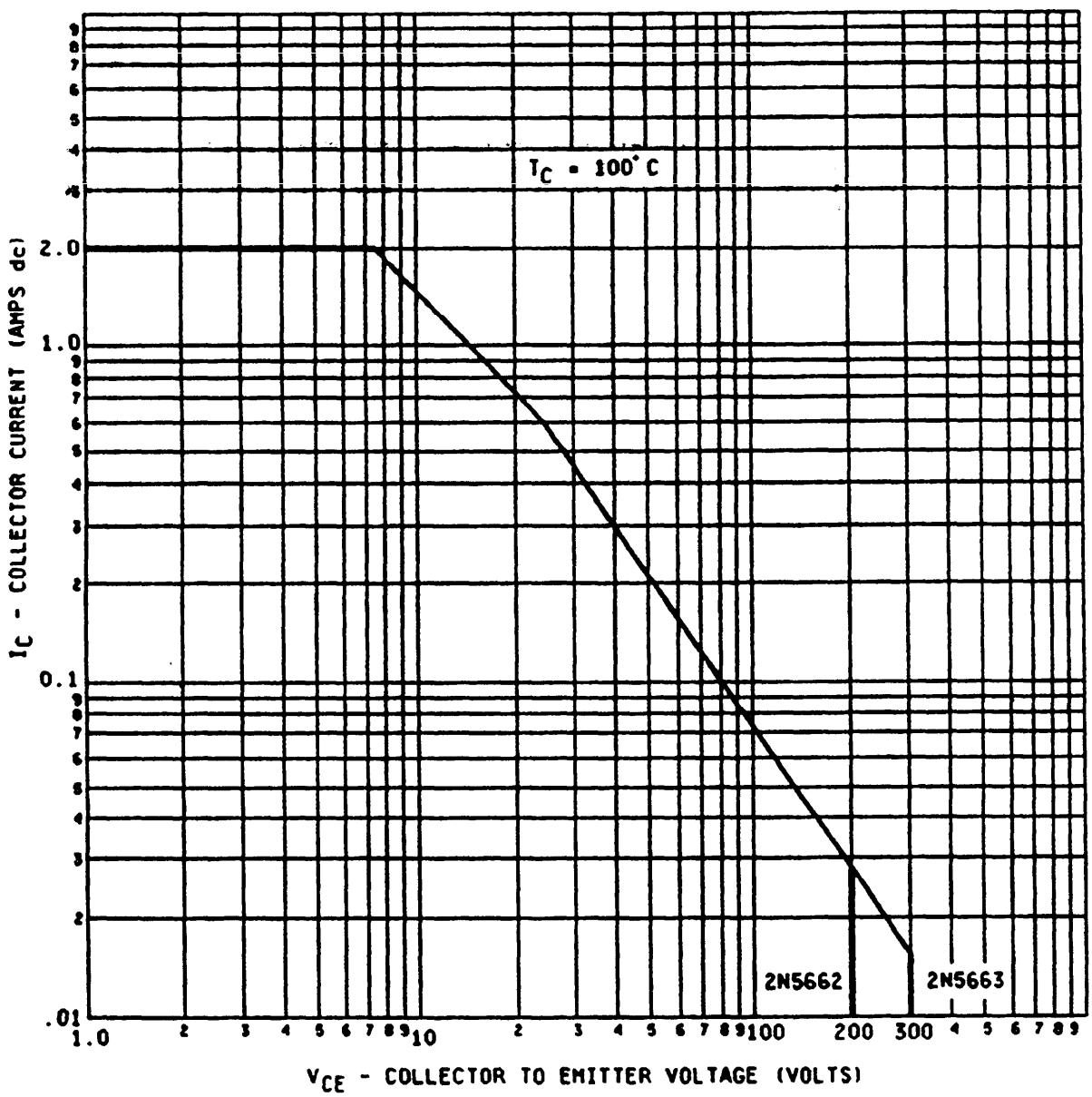



FIGURE 7. Maximum operating area graph (continuous dc) for types 2N5662 and 2N5663.



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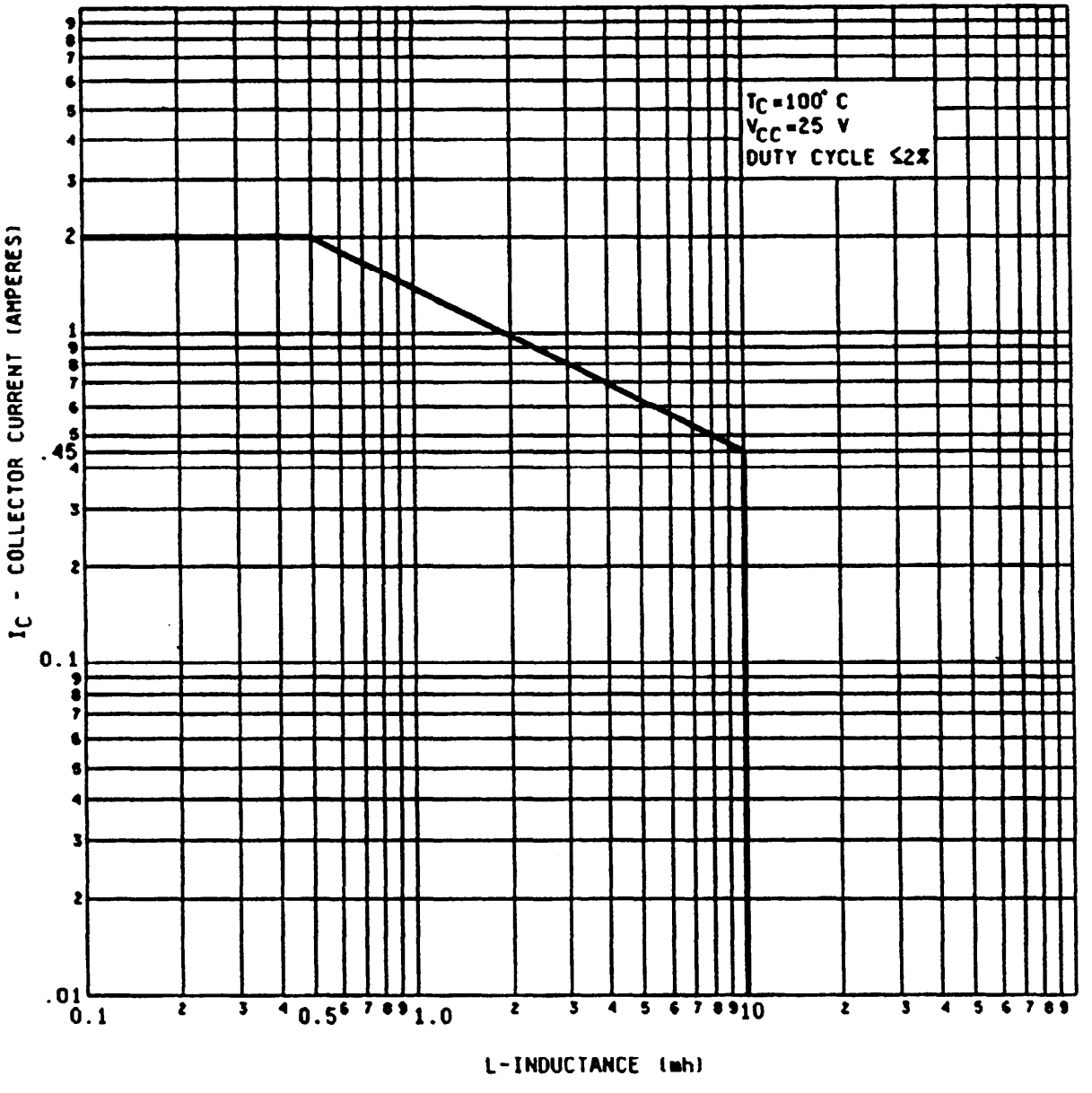



FIGURE 8. Safe operating area for switching between saturation and cutoff - (unclamped inductive load).



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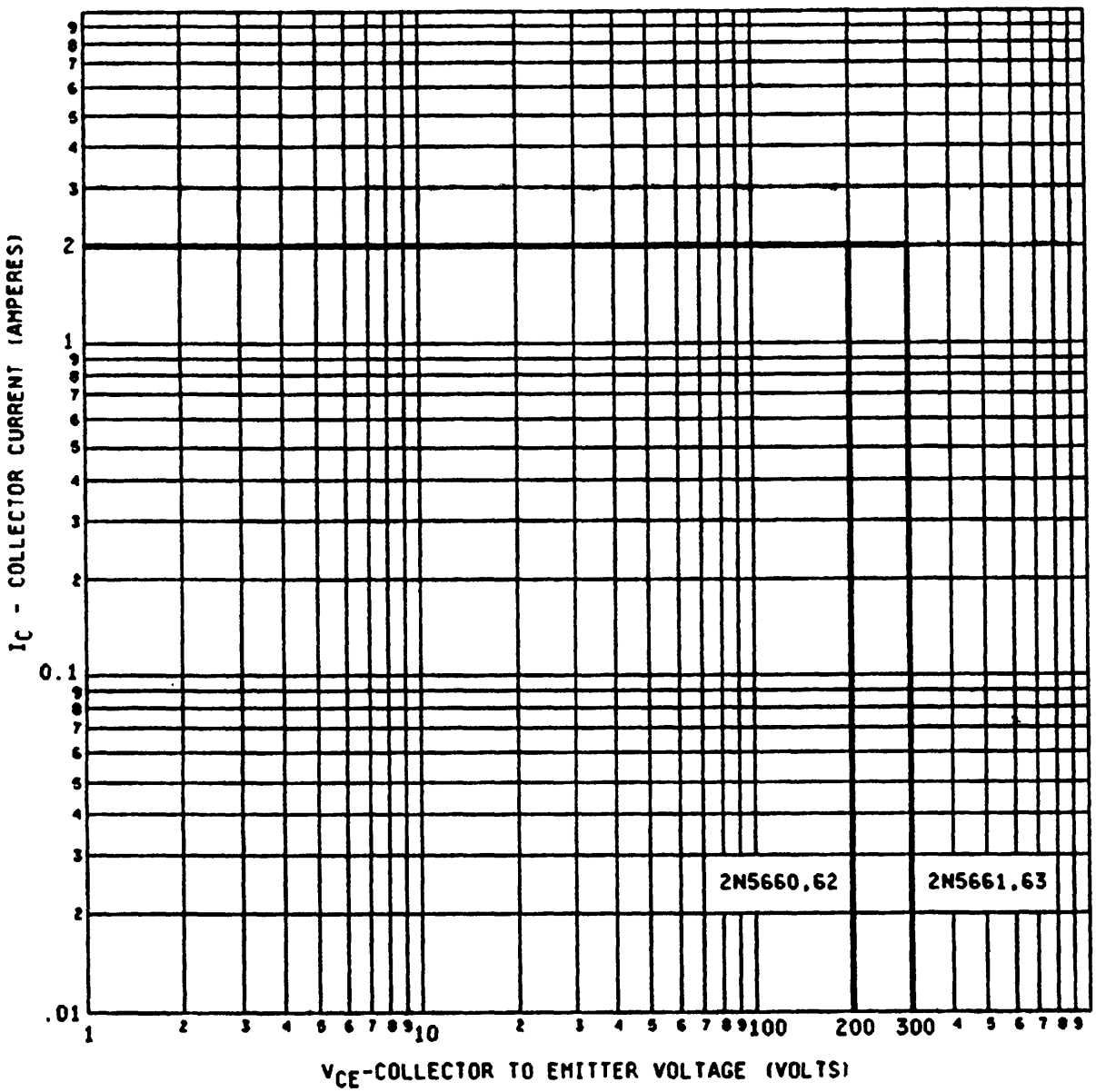
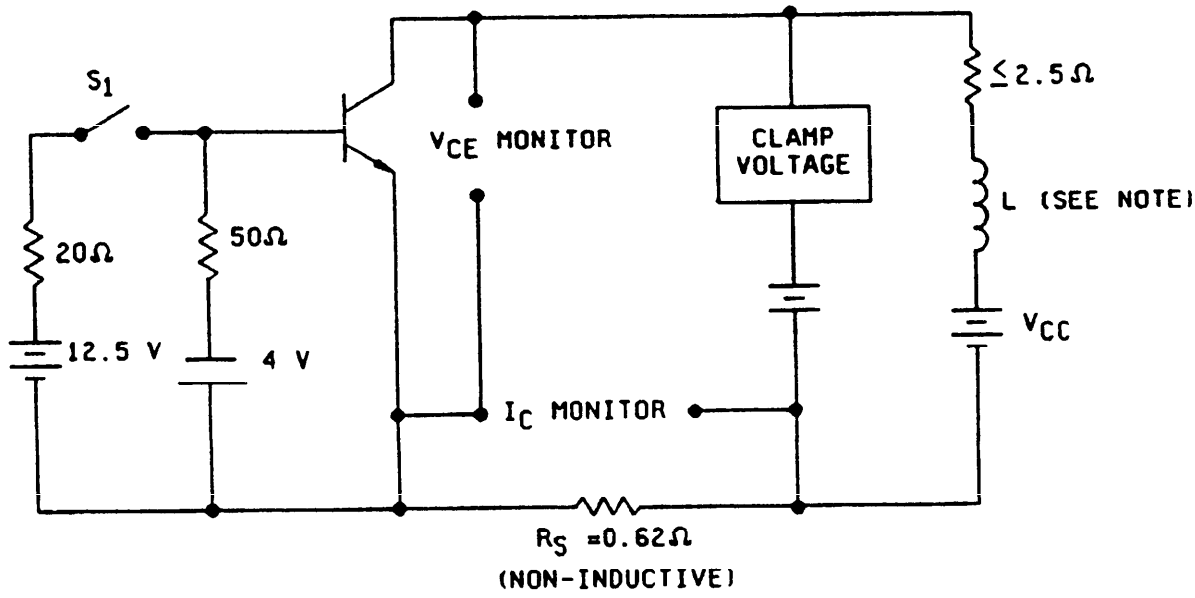



FIGURE 9. Safe operating area for switching between saturation and cutoff - (clamped inductive load).



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NOTE: 40 mH (triad C-48U, or equivalent)

Procedure:

1. With switch $S_1$ closed, set the specified test conditions.
2. Open $S_1$. Device fails if clamp voltage is not reached.
3. Perform specified end-point tests.

FIGURE 10. Clamped inductive sweep test circuit.



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4.5.2 Thermal resistance. Thermal resistance measurements shall be conducted in accordance with method 3131, MIL-STD-750. The following details shall apply:

a. Collector current magnitude during power application shall be 0.5 A dc for 2N5660 and 2N5661, and 0.4 A dc for 2N5662 and 2N5663.

b. Collector to emitter voltage magnitude shall be 20 V dc.

c. Reference temperature measuring point shall be the case.

d. Reference point temperature shall be $+25°C \leq T_R \leq +75°C$ and recorded before the test is started.

e. Mounting arrangement shall be with heat sink to header.

f. Maximum limit of $R_{\theta JC}$ shall be 5°C/W for 2N5660 and 2N5661, and 6.67°C/W for 2N5662 and 2N5663.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Issue of DODISS to be cited in the solicitation.

b. Lead finish as specified (see 3.3.1).

c. Type designation and product assurance level.

6.3 Substitution information. Devices covered by this specification are substitutable for the manufacturer's and user's PIN. This information in no way implies that manufacturer's PIN's are suitable as a substitute for the military Part or Identifying Number (PIN). The term PIN is equivalent to the term (part number, identification number, and type designator) which was previously used in this specification.

6.3.1 JANTX substitution. JANTX devices are a direct replacement for JAN marked devices (Example: Substitute JANTX2N5660 for JAN2N5660).

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.



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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, MI, SM
Navy - AS, CG, MC, OS
Air Force - 13, 19, 85, 99
DLA - ES

Preparing activity:
NASA - NA

Agent:
DLA - ES

(Project 5961-1395)



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| I RECOMMEND A CHANGE: | 1. DOCUMENT NUMBER<br>MIL-S-19500/454B | 2. DOCUMENT DATE (YYMMDD)<br>931101 |
|---|---|---|

3. DOCUMENT TITLE SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, POWER TYPES 2N5660, 2N5661, 2N5662 AND 2N5663 JANTX AND JANTXV

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

| a. NAME (Last, First, Middle initial) | b. ORGANIZATION | |
|---|---|---|
| c. ADDRESS (Include Zip Code) | d. TELEPHONE (Include Area Code)<br>(1) Commercial<br>(2) AUTOVON<br>(If applicable) | 7. DATE SUBMITTED<br>(YYMMDD) |

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| a. NAME | b. TELEPHONE (Include Area Code)<br>(1) Commercial (2) AUTOVON |
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